MAT 319 HW05

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Problem 15.1. *Determine which of the following series converge.*

(a) $\sum \frac{(-1)^n}{n}$

(b) $\sum \frac{(-1)^n n!}{2^n}$

Solution. The first series converges. Since $a_n = \frac{1}{n}$ is a nonnegative, decreasing sequence with limit zero, the alternating series theorem shows us that the series

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$$

converges. We can pull out a constant term of -1 to see that

$$\sum_{n=1}^{\infty}\frac{(-1)^n}{n}$$

also converges, as desired.

The second series does not converge. To prove this, we show that the sequence $b_n = \frac{(-1)^n n!}{2^n}$ does not converge to zero.

> **Claim —** For all natural numbers n, $|b_n| \geq \frac{1}{2}$.

Proof. We induct on n, with the base case $n = 1$ giving $|b_n| = |-\frac{1}{2}| \geq \frac{1}{2}$.

In general, if the claim holds for n, then $n + 1 \geq 2$, so the chain of inequalities

$$|b_{n+1}| = \left|\frac{(-1)^{n+1}(n+1)!}{2^{n+1}}\right| = \left|\frac{-1\cdot(n+1)}{2}\cdot\frac{(-1)^n n!}{2^n}\right| = \left|\frac{-(n+1)}{2}\right|\cdot|b_n| \geq \frac{2}{2}\cdot|b_n| = |b_n|$$

implies that the claim then holds for $n + 1$, too. □

In order for $\lim b_n = 0$ to hold, we need $|b_n - 0| = |b|_n < \varepsilon$ eventually for any given positive real ε; however, the claim above shows that the above statement fails when $\varepsilon = \frac{1}{2}$. However, $\sum b_n$ converging would imply $\lim b_n = 0$, so $\sum b_n$ cannot converge. □

Problem 15.3. *Show that* $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ *converges if and only if* $p > 1$.

Solution 1. Since we haven't properly defined integration yet (or logs, but I can't avoid those), I decided to find a solution without integration.

By factoring out the constant $\frac{1}{(\log 2)^p}$, we see that it is equivalent to determine when

$$\sum_{n=2}^{\infty} \frac{1}{n(\frac{\log n}{\log 2})^p} = \sum_{n=2}^{\infty} \frac{1}{n(\log_2 n)^p}$$

converges. Let $a_n = \frac{1}{n(\log_2 n)^p}$.

First, suppose $p > 1$. For $n \geq 2$, define the sequence

$$b_n = \frac{1}{2^r \cdot r^p}$$

where r is the integer such that $2^r \leq n < 2^{r+1}$. Clearly $r \leq \log_2 n$, so we can say that

$$a_n = \frac{1}{n} \cdot \frac{1}{(\log_2 n)^p} \leq \frac{1}{2^r \cdot r^p} = b_n.$$

Since a_n and b_n are both nonnegative and $a_n \leq b_n$ for all $n \geq 2$, the comparison test says that the convergence of $\sum_{n\geq 2} a_n$ follows from the convergence of $\sum_{n\geq 2} b_n$, so we now prove the convergence of $\sum_{n\geq 2} b_n$.

Since $b_n > 0$ for all n, the sequence (s_n) of partial sums of (b_n) has a limit L, which may be either a positive real number or $+\infty$. Moreover, any subsequence of (s_n) converges to L, so consider the subsequence $(s_{n_k})_{k\geq 2}$ given by $n_k = 2^k - 1$ for integers $k \geq 2$. We can compute

$$\begin{aligned} s_{2^k-1} &= \sum_{j=2}^{2^k-1} b_j = \sum_{r=1}^{k-1} \sum_{j=2^r}^{2^{r+1}-1} b_j \\ &= \sum_{r=1}^{k-1} \sum_{j=2^r}^{2^{r+1}-1} \frac{1}{2^r \cdot r^p} = \sum_{r=1}^{k-1} \frac{1}{2^r \cdot r^p} \cdot 2^r \\ &= \sum_{r=1}^{k-1} \frac{1}{r^p}. \end{aligned}$$

The first line comes from partitioning the set $\{2, 3, \ldots, 2^k - 1\}$ into subsets of the form $\{2^r, 2^r + 1, \ldots, 2^{r+1} - 1\}$; that is, the set corresponding to r is the set of integers x satisfying $2^r \leq x < 2^{r+1}$, and each $x \in \{2, 3, \ldots, 2^k - 1\}$ satisfies $2^r \leq x < 2^{r+1}$ for exactly one integer r between 1 and $k - 1$ inclusive, so each x is in exactly one set of the partition. Thus, $(s_{n_k})_{k\geq 2}$ is exactly the sequence of partial sums of the series

$$\sum_{n\geq 1} \frac{1}{n^p},$$

which, for $p > 1$ is known to converge! By the earlier comparison argument, this means $\sum a_n$ also converges.

It remains to prove the converse, that is, when $p \geq 1$, $\sum a_n$ does not converge. We show that it diverges to $+\infty$ with a similar comparison test argument. This time, for $n \geq 2$, define the sequence

$$c_n = \frac{1}{2^r \cdot r^p}$$

where r is the integer such that $2^{r-1} < n \le 2^r$. We see that $\log_2 n \le r$, so

$$a_n = \frac{1}{n} \cdot \frac{1}{(\log_2 n)^p} \ge \frac{1}{2^r \cdot r^p} = c_n.$$

Since c_n is always nonnegative and $a_n \ge c_n$ for $n \ge 2$, the comparison test says that $\sum_{n\ge 2} a_n = +\infty$ follows from $\sum_{n\ge 2} c_n = +\infty$.

Again, c_n being nonnegative over all n means that the partial sums (t_n) of (c_n) have a limit ℓ, so any subsequence of (t_n) also has limit ℓ. To show that $\ell = +\infty$, we consider the subsequence $(t_{m_k})_{k\ge 2}$ given by $m_k = 2^k$:

$$\begin{aligned} s_{2^k} &= \sum_{j=2}^{2^k} c_j = \sum_{r=1}^{k} \sum_{j=2^{r-1}+1}^{2^r} c_j \\ &= \sum_{r=1}^{k} \sum_{j=2^{r-1}+1}^{2^r} \frac{1}{2^r \cdot r^p} = \sum_{r=1}^{k} \frac{1}{2^r \cdot r^p} \cdot \frac{1}{2^{r-1}} \\ &= \sum_{r=1}^{k} \frac{1}{2r^p}. \end{aligned}$$

This subsequence is thus half the partial sums of the series $\sum \frac{1}{n^p}$. Since, for $p \le 1$, $\sum \frac{1}{n^p}$ diverges to $+\infty$, the scaled version $\sum \frac{1}{2r^p}$ also diverges to $+\infty$. Thus, $\sum_{n\ge 2} c_n$ also diverges to $+\infty$, and at this point the conditions of the comparison test are fulfilled, so $\sum_{n\ge 2} a_n = +\infty$ is not a convergent series. $\square$

Remark. This solution relies on knowing which values of p make $\sum \frac{1}{n^p}$ converge, which we proved using integration; however, we can prove this without integration as well by a similar argument with bounding via powers of two (analogous to our proof of divergence of $\sum \frac{1}{n}$).

Solution 2. Just to show I know how the integral test works. Since $f(x) = \frac{1}{x(\log x)^p}$ is decreasing as x increases (because x and $(\log x)^p$ are both increasing in x), we can get the bounds

$$\int_2^N f(x)\,dx \le \sum_{n=2}^{N} f(n) \le f(2) + \int_3^N f(x-1)\,dx = f(2) + \int_2^N f(x)\,dx$$

for any integer $N > 2$. If $\int_2^N f(x)\,dx$ diverges as N goes to infinity, then $\sum_{n\ge 2} f(n)$, being bounded below by $\int_2^\infty f(x)\,dx$, must also diverge. Similarly, if the integral converges, then $\sum_{n\ge 2} f(n)$ is bounded above by $f(2) + \int_2^\infty f(x)\,dx$, so the partial sums are increasing (because $f(n) > 0$ for $n \ge 2$) and bounded, hence the series converges. In other words, the convergence of the series is equivalent to the convergence of $\int_2^\infty f(x)\,dx$.

Now, we compute the integral to determine when it converges. Let $u = \log x$ so that $du = \frac{dx}{x}$:

$$\begin{aligned} \int_2^\infty \frac{1}{x(\log x)^p}\,dx &= \lim_{N\to\infty} \int_2^N \frac{1}{x(\log x)^p}\,dx \\ &= \lim_{N\to\infty} \int_{\log 2}^{\log N} \frac{1}{u^p}\,du \\ &= \lim_{N\to\infty} \begin{cases} \frac{(\log N)^{1-p}}{1-p} - \frac{(\log 2)^{1-p}}{1-p} & \text{if } p \ne 1 \\ \log\log N - \log\log 2 & \text{if } p = 1 \end{cases} \end{aligned}$$

Since $\log N \to \infty$ as $N \to \infty$, if $1 - p > 0$, $(\log N)^{1-p}$ grows arbitrarily large (so the integral diverges), while if $1 - p < 0$, then $(\log N)^{1-p}$ grows arbitrarily small (so the integral converges). Furthermore, if $p = 1$, then $\log \log N$ also goes to infinity. Thus, the integral converges if and only if $p > 1$, which means the series also converges if and only if $p > 1$. $\square$

Problem 15.6a. *Give an example of a divergent series $\sum a_n$ for which $\sum a_n^2$ converges.*

Solution. Let $a_n = \frac{1}{n}$ for all natural n. Then we have already shown that $\sum \frac{1}{n}$ diverges to $+\infty$, while $\sum \frac{1}{n^2}$ converges. (In fact $a_n = \frac{1}{n^p}$ for any $p \in (\frac{1}{2}, 1]$ easily does the trick). $\square$

Problem 15.7a. *Prove that if (a_n) is a decreasing sequence of real numbers and if $\sum a_n$ converges, then $\lim na_n = 0$.*

Solution. Pick any real number $\varepsilon > 0$; we will show that $|na_n - 0| < \varepsilon$ is eventually true. To start with, we note that $\sum a_n$ converging means that it satisfies the Cauchy criterion, i.e.

$$\text{there exists an } N \text{ such that } n > m > N \implies \left|\sum_{k=m}^{n} a_k\right| < \frac{\varepsilon}{2}.$$

Claim — All terms of the sequence (a_n) are nonnegative.

Proof. Because $\sum a_n$ converges, $\lim_{n\to\infty} a_n = 0$. Since a_n is a decreasing sequence, it converges to $\inf\{a_n\}$, which is thus zero; this implies that $a_n \geq 0$ for all n, as $\inf\{a_n\} = 0$ is, by definition, less than or equal to each of the a_ns. $\square$

Now, since $a_k \leq a_n$ whenever $k \geq n$, we can say that

$$(n - N)a_n = \sum_{k=N+1}^{n} a_n \leq \sum_{k=N+1}^{n} a_k = \left|\sum_{k=N+1}^{n} a_k\right| < \frac{\varepsilon}{2}$$

whenever $n > N$. To be clear, the left inequality is from (a_n) being decreasing, the equality of sums is from (a_n) always being nonnegative, and the right inequality comes from the Cauchy criterion because $n > N + 1 > N$.

Since N is just a constant, we can say that

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} Na_n = 0,$$

so eventually $Na_n < \frac{\varepsilon}{2}$ as well, say when $n > M$. This means that

$$n > \max(M, N) \implies |na_n - 0| = na_n = (n - N)a_n + Na_n < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus, $|na_n - 0|$ is eventually smaller than any positive real number, so $\lim_{n\to\infty} na_n = 0$. $\square$

Problem 17.2. *Let $f(x) = 4$ for $x \geq 0$, $f(x) = 0$ for $x < 0$, and $g(x) = x^2$ for all x. Thus* $\mathrm{dom}(f) = \mathrm{dom}(g) = \mathbb{R}$.

(a) *Determine the following functions: $f + g$, fg, $f \circ g$, $g \circ f$. Be sure to specify their domains.*

(b) *Which of the functions f, g, $f + g$, fg, $f \circ g$, $g \circ f$ is continuous?*

Solution. For (a):

- $(f + g)(x) = x^2$ for $x < 0$ and $4 + x^2$ for $x \geq 0$.
- $(fg)(x) = 0 \cdot x^2 = 0$ for $x < 0$ and $4x^2$ for $x \geq 0$.
- $(f \circ g)(x) = f(0) = 0$ for $x < 0$ and $f(4) = 16$ for $x \geq 0$
- $(g \circ f)(x) = g(x^2) = 4$ for all x because $x^2 \geq 0$ for all real x.

For (b), I claim that g, fg, and $g \circ f$ are continuous while the rest of the functions aren't. First, to see that these are continuous:

- $g(x)$ is continuous because $\lim_{n\to\infty} x_n = x$ implies
$$\lim_{n\to\infty} x_n^2 = \lim_{n\to\infty} x_n \cdot \lim_{n\to\infty} x_n = x \cdot x = x^2.$$
- We show that $(fg)(x)$ is continuous at r by considering three cases based on the sign of r. If $r > 0$, then any sequence (x_n) converging to r is eventually positive (since $|x_n - r| < |r|$ eventually). This means $f(x_n) = 4x_n^2$ is eventually true, so
$$\lim_{n\to\infty} (fg)(x_n) = \lim_{n\to\infty} 4x_n^2 = 4r^2 = (fg)(r).$$
Similarly, if $r < 0$, then any sequence (x_n) converging to r is eventually negative and thus $(fg)(x_n)$ is eventually always zero; thus in this case
$$\lim_{n\to\infty} (fg)(x_n) = \lim_{n\to\infty} 0 = 0 = (fg)(r).$$
Finally, if $r = 0$, then observe that, for all $\varepsilon > 0$,
$$|x - r| = |x - 0| < \frac{\sqrt{\varepsilon}}{2} \implies |(fg)(x) - (fg)(r)| = |(fg)(x)| < \varepsilon$$
because, if $x \geq 0$, then $|(fg)(x)| = 4x^2 < 4(\frac{\sqrt{\varepsilon}}{2})^2 = \varepsilon$, and, if $x < 0$, then $|(fg)(x)| = 0 < \varepsilon$, so fg is continuous at 0 by the ε-δ definition of continuity (with $\delta = \sqrt{\varepsilon}$ here).
- Finally, $g \circ f$ is constantly 4, so it is clearly continuous at any real r, since, for any sequence (x_n) with limit r,
$$\lim_{n\to\infty} (g \circ f)(x_n) = \lim_{n\to\infty} 4 = 4 = (g \circ f)(r).$$

Now, we show that the other three functions are all discontinuous at 0. In each case, we need to find a sequence of real numbers (x_n) that converges to zero such that the sequence $(f(x_n))$ doesn't converge to $f(0)$. We show that taking $x_n = -\frac{1}{n}$ suffices; the important property is that $x_n < 0$ for all n.

- For f, observe that $f(x_n) = 0$ because $x_n < 0$ for all n, so
$$\lim_{n\to\infty} f(x_n) = 0 \neq 4 = f(0).$$
- For $f+g$, $(f+g)(x_n) = x_n^2$ because $x_n < 0$ for all n, so
$$\lim_{n\to\infty} (f+g)(x_n) = (\lim_{n\to\infty} x_n)^2 = 0 \neq 4 = f(0).$$
- For $f \circ g$, $(f \circ g)(x_n) = 0$ because $x_n < 0$ for all n, so
$$\lim_{n\to\infty} (f \circ g)(x_n) = 0 \neq 16 = f(0).$$

This is enough to prove that these three functions are all not continuous at zero, hence not continuous. □

Problem 17.4. *Prove the function* $\sqrt{x}$ *is continuous on its domain* $[0, \infty)$.

Solution. To prove that $f(x) = \sqrt{x}$ is continuous at any $r \in [0, \infty)$, we want to prove that, for each $\varepsilon > 0$, there exists δ such that
$$|x - r| < \delta \implies |\sqrt{x} - \sqrt{r}| = |f(x) - f(r)| < \varepsilon.$$

If $r > 0$, then, taking $\delta = \varepsilon\sqrt{r}$, we can see by difference of squares that
$$\left|\sqrt{x} - \sqrt{r}\right| = \frac{|x - r|}{\sqrt{x} + \sqrt{r}} < \frac{\delta}{\sqrt{x} + \sqrt{r}} < \frac{\delta}{\sqrt{r}} = \varepsilon.$$

If $r = 0$, then take $\delta = \varepsilon^2$. Note that $f(x)$ is increasing because, if $a = u^2$ and $b = v^2$ where $u = \sqrt{a} = f(a)$ and $v = \sqrt{b} = f(b)$ are real numbers,
$$a \leq b \iff u^2 - v^2 = (u + v)(u - v) \leq 0,$$
which necessitates $u - v \leq 0 \iff u \leq v$ because $u + v \geq 0$. Thus, we have
$$x = |x - 0| < \delta = \varepsilon^2 \implies \sqrt{x} = |\sqrt{x} - 0| = |f(x) - f(0)| < \sqrt{\delta} = \varepsilon,$$
as desired. □

Problem 17.9c. *Prove that* $f(x)$*, defined as* $f(r) = r\sin(\frac{1}{r})$ *for* $r \neq 0$ *and* $f(0) = 0$*, is continuous at* 0.

Solution. For any $\varepsilon > 0$, let $\delta = \varepsilon$. Then, to prove that f is continuous at zero, we need to show that, for every $\varepsilon = 0$,
$$|x| = |x - 0| < \delta = \varepsilon \implies |f(x)| = |f(x) - f(0)| < \varepsilon.$$

Indeed, if $|x| < \delta$, then
$$|f(x)| = \left|x \sin\left(\frac{1}{x}\right)\right| = |x| \cdot \left|\sin\left(\frac{1}{x}\right)\right| \leq |x| \cdot 1 < \delta = \varepsilon,$$
so the desired implication holds and $f(x)$ is continuous at 0. □

Problem 17.13a. *Let $f(x) = 1$ when x is rational and $f(x) = 0$ when x is irrational. Show f is discontinuous at every x in $\mathbb{R}$.*

Solution. Let $r \in \mathbb{R}$ be any real number. To show that f is discontinuous at r, we need to show that there exists an $\varepsilon > 0$ such that, regardless of the value of $\delta > 0$, the implication

$$|x - r| < \delta \implies |f(x) - f(r)| < \varepsilon$$

does not hold.

Pick $\varepsilon = 1$. Then, we know that, for every pair of real numbers a and b with $a < b$, there exists a rational number in (a, b) and an irrational number in (a, b) (the former is just a result of denseness of $\mathbb{Q}$ in $\mathbb{R}$, and the latter was proven in a homework). In this case, taking $a = r - \delta$ and $b = r + \delta$, we see that there is a rational number $q \in (r - \delta, r + \delta)$ and an irrational number $y \in (r - \delta, r + \delta)$. By definition, $|q - r| < \delta$ and $|y - r| < \delta$, and $f(q) = 1$ and $f(y) = 0$. Thus, either $f(r) \neq f(q)$ or $f(r) \neq f(y)$. If $f(r) \neq f(q) = 1$, then $f(r) = 0$, so we get

$$|f(r) - f(q)| = |0 - 1| = 1 \geq 1 = \varepsilon,$$

and if $f(r) \neq f(y) = 0$, then $f(r) = 1$, so similarly $|f(r) - f(y)| = 1 \geq \varepsilon$. Thus, we have shown that the implication

$$|x - r| < \delta \implies |f(x) - f(r)| < \varepsilon$$

fails for $\varepsilon = 1$, which is enough to prove that f is discontinuous at any real number r. $\square$